

The determinant as an orientation invariant and the source of the cubic term in equivariant matrix gradient flows

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DOI: 10.5281/zenodo.20752208 (v3)

Self-contained manuscript in the language of dynamical systems; it requires no external framework. Spanish version: determinante_cubico_flujos_gradiente_molina2024.md. Reproducible verification scripts, one per numerical statement (§8), are at

https://github.com/hmolina/papers/tree/main/determinant_cubic_source/code

The inertial extension (Bogdanov–Takens) and the phenomenology of chaos are treated, as numerically explored directions and not as theorems, in the companion document outlook_inercial_caos.md.

Abstract

We consider the gradient flow $\dot{\Gamma} = -\nabla P(\Gamma; \mu, J)$ on the real 4×4 matrices, with $P = \|\Gamma\|^2 + \mu \det \Gamma + \beta \|\Gamma\|^4 + b_6 \|\Gamma\|^6 - \langle J, \Gamma \rangle$, and its damped second-order version. The field-free potential is invariant under the bilateral orthogonal action $\Gamma \mapsto U\Gamma V^\top$ ($U, V \in O(4)$, $\det U \det V = 1$). The ring of invariants of this action is generated by the symmetric functions of the singular values and by the determinant; the latter is the only orientation-sensitive generator, and the naive candidate for a cubic invariant, $\text{tr } \Gamma^3$, is not invariant under this symmetry. Near a Hessian degeneracy—a simple soft mode, isolated by the linear term J in the sense of imperfect bifurcation—centre-manifold reduction yields the local normal forms, and the cubic coefficient receives from the determinant, through its cofactor matrix, its unique orientation-sensitive contribution; within the potential considered it is its only source on the stratum $V \perp \Gamma_*$. We prove the codimension-1 reduction (fold and, on strata with $\mathbb{Z}_2$ isotropy, pitchfork) with explicit coefficients, and the codimension-2 cusp A_3 . A no-Hopf lemma shows that no metric-gradient flow, for any positive-definite metric, admits an oscillatory bifurcation: the spectrum of the degeneracy is real. The same mechanism, under the conjugation symmetry $O(n)$ of symmetric tensors, selects $\text{tr } Q^3$, as in the Landau–de Gennes energy of the nematic Q -tensor. Each statement is accompanied by reproducible numerical verification.

Keywords: equivariant bifurcation, matrix gradient flow, centre-manifold reduction, normal forms, cofactor matrix, orientation invariant, no-Hopf lemma, cusp.

1. Introduction

Bifurcation theory organizes the qualitative changes of a dynamical system around a catalogue of normal forms (Guckenheimer and Holmes 1983; Kuznetsov 2004). When the field has a symmetry, equivariant theory (Golubitsky, Stewart and Schaeffer 1988) determines which bifurcations are generic, why degeneracies appear in families, and which invariants may enter the normal form. We study, for a gradient flow with a matrix variable and bilateral orthogonal symmetry, the origin of the cubic term of the soft-mode bifurcation.

The content is that the structure of the invariant ring of the symmetry fixes that term. The determinant is the only generator of the bilateral ring that is not a function of the tensor $\Gamma^\top \Gamma$, that is, the only orientation-sensitive one; upon projecting onto the centre manifold it supplies the orientation part of the cubic. The third-degree invariant in eigenvalues, $\text{tr } \Gamma^3$, does not belong to the bilateral ring. The same logic, for the conjugation symmetry of symmetric tensors, selects $\text{tr } Q^3$: it is the cubic of Landau–de Gennes theory (§7). The result of this work is the bilateral instance of that pattern.

We prove the codimension-1 reduction with explicit coefficients (§3–§4), classify the codimension-2 cusp (§5), establish a no-Hopf lemma that delimits the scope of the gradient sector (§6), place the mechanism alongside its conjugation counterpart (§7), and verify each result numerically (§8).

2. The system and its symmetry

Let $V = M_4(\mathbb{R}) \cong \mathbb{R}^{16}$ with the Frobenius inner product $\langle X, Y \rangle = \text{tr}(X^\top Y)$. We take

$$P(\Gamma; \mu, J) = \|\Gamma\|^2 + \mu \det \Gamma + \beta \|\Gamma\|^4 + b_6 \|\Gamma\|^6 - \langle J, \Gamma \rangle, \quad \beta, b_6 \geq 0,$$

with control parameter μ and external field $J \in M_4(\mathbb{R})$. The sextic b_6 does not enter the local statements; it stabilizes globally the subcritical branches when the slaving correction (Theorem 1.4) turns the effective quartic negative.

For $J = 0$, P is invariant under $\Gamma \mapsto U\Gamma V^\top$ of the group $G = \{(U, V) \in O(4) \times O(4) : \det U \det V = 1\}$, since $\|U\Gamma V^\top\|^2 = \|\Gamma\|^2$ and $\det(U\Gamma V^\top) = \det \Gamma$. This is the singular-value-decomposition symmetry.

The polynomial invariants of G are functions of the singular values σ_i of Γ : they are generated by the power sums $p_k = \text{tr}((\Gamma^\top \Gamma)^k) = \sum_i \sigma_i^{2k}$, all of even degree and functions of $\Gamma^\top \Gamma$, and by the determinant $\det \Gamma = \pm \prod_i \sigma_i$, with $\det^2$ already expressible in the p_k . The determinant is thus the only generator that is not a function of $\Gamma^\top \Gamma$; it is the orientation-sensitive invariant. The naive candidate for a cubic, $\text{tr} \Gamma^3 = \sum_i \lambda_i^3$ (powers of the eigenvalues), is not G -invariant: under $(U, V) \in G$, $\text{tr}(U\Gamma V^\top)^3 = \text{tr}(U\Gamma(V^\top U)\Gamma(V^\top U)\Gamma V^\top)$, which depends on $V^\top U$ and reduces to $\text{tr} \Gamma^3$ only when $V = U$. It lies outside the bilateral ring. Only under the conjugation subsymmetry $U = V$ would it be invariant, which is the case of symmetric tensors (§7).

The continuous symmetry would make the degeneracies appear in G -orbits rather than in isolation. The linear term $-\langle J, \Gamma \rangle$ breaks G . Generically it breaks it completely: it intersects the G -orbits transversally (Thom transversality), lifts the degeneracy, and leaves a single zero eigenvalue of the Hessian (hypothesis H2). This is the theory of imperfect bifurcations (Golubitsky and Schaeffer 1985). Since J enters linearly, it displaces the equilibrium without altering the Hessian. If J preserves an isotropy subgroup $\mathbb{Z}_2 \subset G$ (a field on a symmetric stratum), the soft mode inherits that isotropy and the generic bifurcation is the pitchfork (§3); pitchforks live on the symmetric strata, as is standard in equivariant bifurcation theory.

The gradient flow is

$$\dot{\Gamma} = -\nabla P, \quad \nabla P = 2\Gamma + \mu C(\Gamma) + (4\beta \|\Gamma\|^2 + 6b_6 \|\Gamma\|^4)\Gamma,$$

with $C(\Gamma) = \text{cof}(\Gamma) = \partial \det \Gamma / \partial \Gamma$. The Hessian $H(\Gamma) = D^2 P$ is symmetric. We also consider the damped second-order version $\ddot{\Gamma} + \gamma \dot{\Gamma} + \nabla P = 0$, whose limit $\gamma \rightarrow \infty$ recovers the gradient flow; only to probe basins and escape rates (§8) do we use the Langevin extension $\dot{\Gamma} = -\nabla P + \sqrt{2D} \eta$.

Remark 2.1 (scope and dimension). Parts 1–2 of Theorem 1 are the centre-manifold / Lyapunov–Schmidt reduction for any real-analytic P under (H1)–(H2); only the identification of the cubic coefficient uses the form of P . The construction requires a positive-definite inner product (the Frobenius one), which bounds P from below and fixes the gradient structure. We present $n = 4$ for concreteness and because there $\det$ has degree 4 and $D^3 \det$ is linear; the mechanism extends to $M_n(\mathbb{R})$ with the bilateral action $O(n) \times O(n)$, where $\det$ has degree n . A Monte Carlo sampling over (β, b_6, J) confirms the persistence of the simple soft mode and of the determinant's contribution (§8).

Remark 2.2 (other fourth-degree invariants). The bilateral ring contains a second degree-four invariant besides $\det$ and $\|\Gamma\|^4 = p_1^2$: namely $p_2 = \text{tr}((\Gamma^\top \Gamma)^2)$. Its cubic contribution is $D^3 p_2[V, V, V] = 24 \langle \Gamma_*, V(V^\top V) \rangle$, which is not a function of $\langle \Gamma_*, V \rangle$ and therefore, unlike that of $\|\Gamma\|^4$, does not vanish in general on $V \perp \Gamma_*$. We do not include p_2

in P : we work with the potential above, where on $V \perp \Gamma_*$ the determinant is the only source of the cubic. The statement that survives with p_2 present is weaker but intrinsic: the determinant is the only orientation-sensitive contribution to the cubic, since p_2 , being a function of $\Gamma^\top \Gamma$, is blind to orientation.

3. Main result (codimension 1)

At (Γ_*, μ_*) we assume: (H1) $\nabla P = 0$; (H2) $H_* = H(\Gamma_*)$ has a simple zero eigenvalue, with unit eigenvector V , and the rest of the spectrum bounded away from zero; (H3) transversality $\tau := \langle V, C(\Gamma_*) \rangle \neq 0$; (H4) non-degeneracy $a_3 := D^3 P(\Gamma_*)[V, V, V] \neq 0$, except on strata with $\mathbb{Z}_2$ isotropy, where $a_3 = 0$ and $a_4^{\text{eff}} \neq 0$.

Theorem 1 (reduction $\Gamma \rightarrow \xi$). Under (H1)–(H2) there exists, in a neighbourhood of (Γ_*, μ_*) , a unique smooth one-dimensional centre manifold $\Gamma = \Gamma_* + \xi V + h(\xi, \mu)$ with $h \in V^\perp$, $h = O(\xi^2, \xi \tilde{\mu})$, on which the flow is gradient, $\dot{\xi} = -\partial_\xi \Phi(\xi, \mu)$. If in addition (H3)–(H4):

1. if $a_3 \neq 0$, the reduced form is the fold $\dot{\xi} = \alpha - \frac{1}{2}a_3\xi^2$, with $\alpha = -\tau(\mu - \mu_*)$;
2. if a $\mathbb{Z}_2$ isotropy forces $a_3 = 0$, the reduced form is the pitchfork $\dot{\xi} = -a_2'\xi - \frac{1}{6}a_4^{\text{eff}}\xi^3$;
3. $a_3 = \mu_* D^3 \det(\Gamma_*)[V, V, V] + 24\beta \langle \Gamma_*, V \rangle$; the second term vanishes on the stratum $V \perp \Gamma_*$ (the set of equilibria Γ_* orthogonal to their soft mode V in the Frobenius inner product, $\langle V, \Gamma_* \rangle = \text{tr}(V^\top \Gamma_*) = 0$), where the determinant is the only source of the cubic within P . The determinant's contribution is, moreover, the only orientation-sensitive one (Remark 2.2);
4. $a_4^{\text{eff}} = D^4 P[V^{\otimes 4}] - 3 \langle D^3 P[V, V], (H_*|_{V^\perp})^{-1} D^3 P[V, V] \rangle$; its sign is fixed by the spectrum of $H_*|_{V^\perp}$.

In the symmetric sector, a_4^{eff} changes sign at $\mu = 16\beta$, the line separating the regime of one equilibrium from that of three (Figure 1).

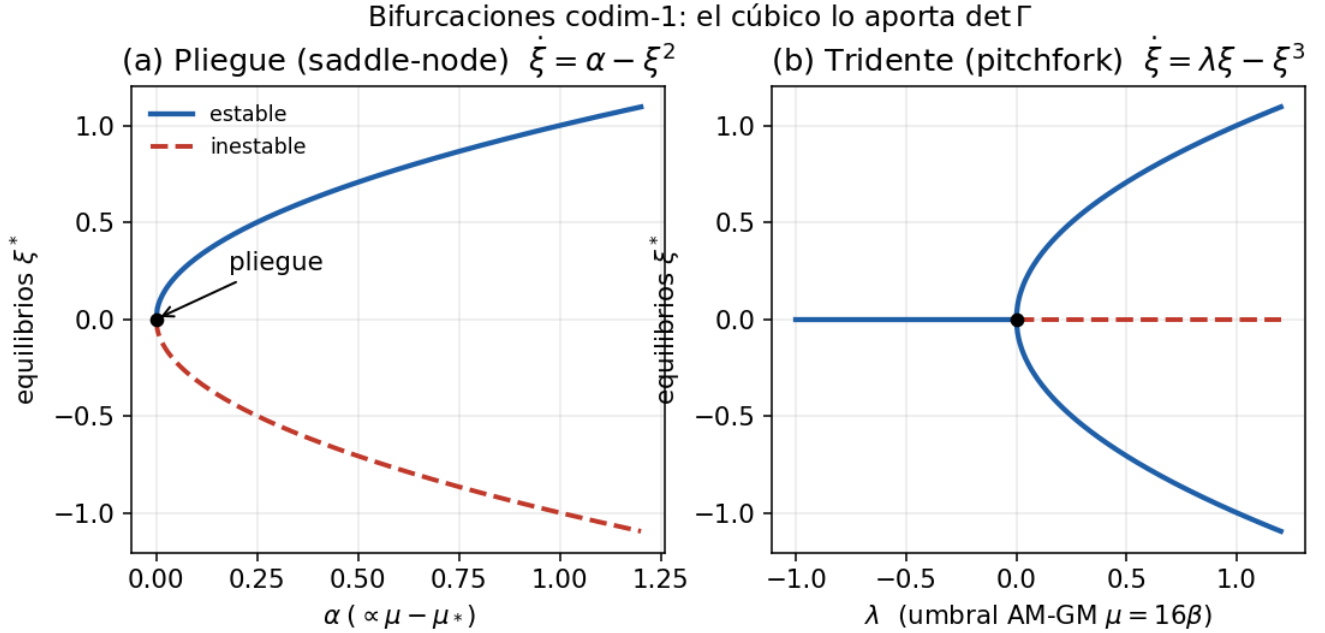


Figure 1: Figure 1. Codimension-1 bifurcations: fold and pitchfork. Stable branches in blue, unstable in dashed red. The cubic term comes from $\det \Gamma$.

4. Proof (Lyapunov–Schmidt)

This is the Lyapunov–Schmidt reduction for gradient fields (Carr 1981; Golubitsky and Schaeffer 1985). With $\Gamma = \Gamma_* + \xi V + W$, $W \in V^\perp$, $\mu = \mu_* + \nu$, and Q the projection onto $V^\perp = \text{ran } H_*$, the component $Q\nabla P = 0$ is solved by the implicit function theorem ($H_*|_{V^\perp}$ invertible by H2) and yields the slaving $W = h(\xi, \mu)$, with $\partial_\xi h(0, \mu_*) = 0$ and $\partial_\xi^2 h(0) = -(H_*|_{V^\perp})^{-1} Q D^3 P(\Gamma_*)[V, V]$. Substituting, the bifurcation equation $g = \langle V, \nabla P \rangle$ coincides, since the field is gradient, with $\partial_\xi \Phi$. With $a_k = \partial_\xi^k \Phi(0, \mu_*)$: $\partial_\nu a_1|_0 = \langle V, C(\Gamma_*) \rangle = \tau$; $a_2(\mu_*) = \langle V, H_* V \rangle = 0$; $a_3 = \langle V, D^3 P[V, V] \rangle$, with no slaving correction since it is $\perp V$.

We decompose $D^3 P$ by terms. The $\|\Gamma\|^2$ term is quadratic and does not contribute. For the quartic norm, $D^3(\|\Gamma\|^4)[V, V, V] = 24\langle \Gamma_*, V \rangle$. For the determinant, of degree 4, $D^3 \det$ is linear and $D^3(\mu \det)[V, V, V] = \mu_* D^3 \det(\Gamma_*)[V, V, V]$. Summing gives point 3. On $V \perp \Gamma_*$ the norm term vanishes and only the determinant term survives.

Explicit Taylor expansion of $\det$. Since $\det$ is multilinear in the four rows, the expansion of $\det(\Gamma_* + \xi V)$ in ξ is exact at degree 4:

$$\det(\Gamma_* + \xi V) = \det \Gamma_* + \xi \langle C(\Gamma_*), V \rangle + \frac{\xi^2}{2} D^2 \det(\Gamma_*)[V, V] + \frac{\xi^3}{6} D^3 \det(\Gamma_*)[V, V, V] + \xi^4 \det V,$$

where $C(\Gamma_*) = \text{adj}(\Gamma_*)$ is the cofactor matrix, and each coefficient picks k rows from V and $4 - k$ rows from Γ_* . In particular, $D^3 \det(\Gamma_*)[V, V, V] = 6 \sum_{j=1}^4 \det(M_j)$, where M_j is the 4×4 matrix whose j -th row is taken from Γ_* and the other three rows from V . This is linear in Γ_* (degree $4 - 3 = 1$), confirming that $D^3 \det$ is a linear polynomial in the equilibrium point, not a function of V alone. The last term $\xi^4 \det V = 0$ whenever V has rank ≤ 3 (e.g., when V is the rank-1 soft mode), so the cubic ξ^3 coefficient is the leading orientation-sensitive term on the centre manifold.

The geometric reading is direct in the singular values: $\det \Gamma = \prod_i \sigma_i$ couples the four irreducibly and supplies the orientation-sensitive part; $\|\Gamma\|^2 = \sum_i \sigma_i^2$ is a function of $\Gamma^\top \Gamma$ and enters only through $\langle \Gamma_*, V \rangle$. The effective quartic follows from differentiating g once more; its sign is fixed by the definiteness of $H_*|_{V^\perp}$. The fold/pitchfork classification follows from Sotomayor’s conditions (1973): with $a_1(\mu_*) = a_2(\mu_*) = 0$ and $a_3 \neq 0$ one obtains the fold; if a $\mathbb{Z}_2$ annuls a_3 , the first nontrivial term is the quartic and one obtains the pitchfork. $\square$

The full symbolic computation is in the repository at https://github.com/hmolina/papers/tree/main/determinant_cubic_source/code and in the technical document `teorema_gamma_xi.md`, §2.

5. Codimension-2 cusp

Theorem 2 (cusp A_3). A point with $a_2 = a_3 = 0$ and $a_4^{\text{eff}} \neq 0$ is a cusp, the universal unfolding of the pitchfork (Thom 1975). Since the contributions of the determinant and of the norm compete, $a_3(\mu, s)$ changes sign and, by the intermediate value theorem, vanishes on a curve; the second control s fixes $a_2 = 0$ at a point of that curve, where $a_4^{\text{eff}} \neq 0$ and the transversality $\partial(a_1, a_2)/\partial(\mu, s)$ is nonsingular, so that the A_3 is versal. The four conditions are certified numerically (§8); the window of three equilibria scales as $(-a_2)^{3/2}$ (Figure 2).

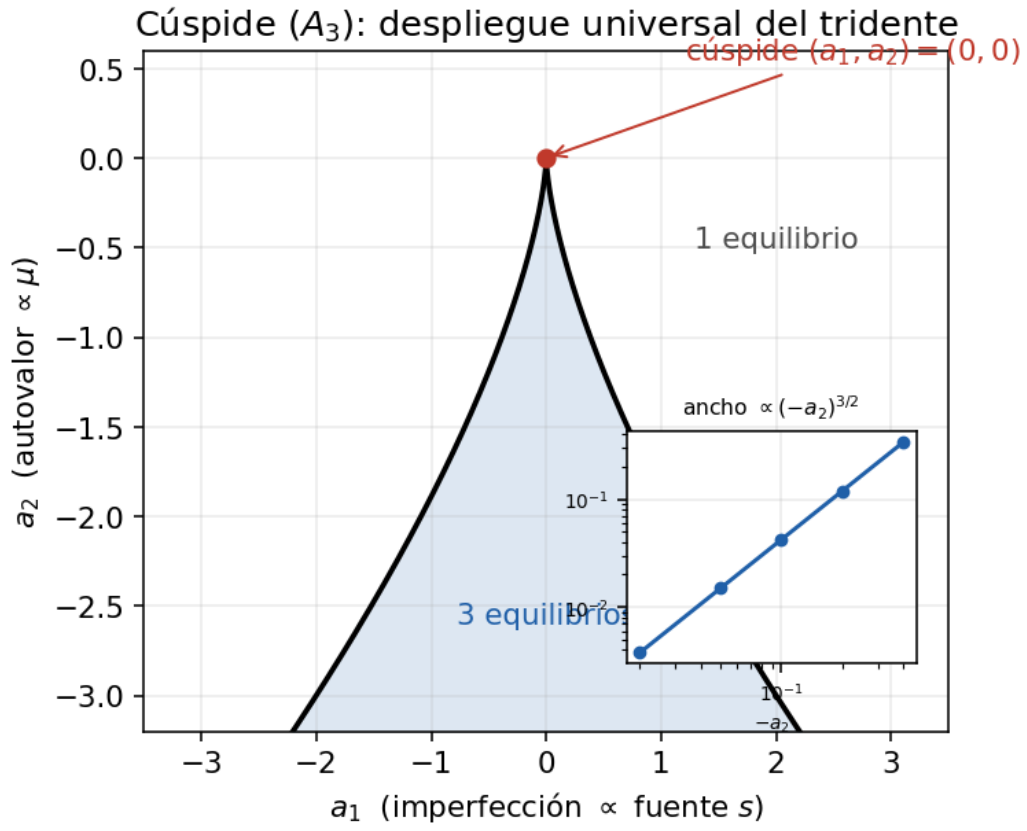


Figure 2: Figure 2. The cusp A_3 in the unfolding plane (a_1, a_2) . Inside the wedge $4a_2^3 + 27a_1^2 \leq 0$ there are three equilibria; outside, one. The inset shows the 3/2 law.

6. A no-Hopf lemma

Lemma 1 (no Hopf in the metric-gradient flow). At an equilibrium Γ_* , the Jacobian of $\dot{\Gamma} = -G^{-1}\nabla P$, for any symmetric positive-definite metric $G \succ 0$ (including a state-dependent Riemannian metric $G(\Gamma)$; at $\nabla P(\Gamma_*) = 0$ the term with the derivative of the metric vanishes), has real spectrum. No complex pair crosses the imaginary axis and no Hopf bifurcation occurs: while the dynamics is metric-gradient, the soft mode admits only stationary bifurcations (fold, pitchfork, cusp). An oscillatory instability requires breaking the gradient form: either G losing positivity, or a non-gradient reactive term.

Proof. The Jacobian at Γ_* is $J = -G_*^{-1}H_*$, with $G_* = G(\Gamma_*) \succ 0$ and $H_* = D^2P(\Gamma_*) \in \text{Sym}$. Write $J = G_*^{-1/2}(-G_*^{-1/2}H_*G_*^{-1/2})G_*^{1/2}$; since $G_*^{-1/2}H_*G_*^{-1/2}$ is symmetric, its spectrum is real, and similarity preserves the spectrum. Hence $\sigma(J) \subset \mathbb{R}$ for every $G_* \succ 0$ and $H_* \in \text{Sym}$. $\square$

The lemma organizes the catalogue by the spectrum of the degeneracy. A simple real eigenvalue crossing zero gives a stationary bifurcation, with the cubic originating from the determinant (Theorems 1–2). An oscillatory regime requires a complex pair crossing the axis, impossible in the gradient sector. The type is read off the critical spectrum.

The lemma admits a thermodynamic reading: strict dissipation, as a gradient flow with a positive metric, is obstructed from producing sustained oscillation or limit cycles. It is the dynamical correlate of why a system in monotone relaxation harbours no self-sustained rhythm. In Onsager’s framework (1931), the purely dissipative sector (symmetric Onsager matrix) admits no oscillatory currents; oscillatory instabilities require the antisymmetric reactive contribution, exactly as Lemma 1 requires a non-gradient term. Unlike Theorems 1–2, the lemma does not depend on the particular potential or on the dimension.

7. The conjugation counterpart: the Q -tensor

The mechanism of §3 has a counterpart under another symmetry. For the conjugation action $Q \mapsto RQR^\top$, $R \in O(n)$, on symmetric tensors, the invariant $\text{tr } Q^3$ is indeed invariant (it is a function of the eigenvalues, which conjugation preserves). In the Landau–de Gennes energy of nematic liquid crystals,

$$F(Q) = \frac{a}{2} \text{tr } Q^2 - \frac{b}{3} \text{tr } Q^3 + \frac{c}{4} (\text{tr } Q^2)^2,$$

it is $\text{tr } Q^3$ that supplies the cubic responsible for the first-order isotropic–nematic transition (de Gennes and Prost 1993; Mottram and Newton 2014). Thus, depending on the symmetry, the lowest-degree invariant that breaks isotropy changes: the determinant for the bilateral action, $\text{tr } Q^3$ for conjugation. The result of this work is the bilateral instance; the conjugation one is known physics. We do not claim a general theorem encompassing both; the observation is that the cubic of the normal form is supplied, in each case, by the lowest-degree anisotropic invariant admitted by the symmetry, and it should be verified case by case. Q -tensors and continuum strain tensors are real matrix gradient flows and offer a testing ground.

8. Numerical verification

Each statement is accompanied by a self-contained, reproducible script (NumPy/SciPy). Degeneracies are located by solving $\nabla P = 0$ and $\lambda_{\min}(H) = 0$; the normal-form coefficients by polynomial fitting; the Lyapunov exponents by Benettin (1980); the escape rates by Kramers (Hänggi et al. 1990); the equilibrium branches by pseudo-arclength continuation (Doedel 1981).

#	object	result	script
1	fold/pitchfork as invariant objects; $\text{Var} \sim 1/k$; Monte Carlo 10^3	critical pitchfork $k_2 = 0$; 100% classification	pieza1_bifurcaciones_rigor.py
2	exact reduction (invariant ray)	$\nabla P \parallel \Gamma$; threshold $\mu = 16\beta$	pieza1_reduccion_normal_forms.py
3	generic centre manifold	full flow = reduced (not the naive one)	pieza1_centro_manifold_generico.py
4	Kramers + continuation	$\ln \langle \tau \rangle = 0.904 \Delta U / D + 1.78$, $R^2 = 0.989$; fold to $\mu \approx 2.04$	pieza1_kramers_continuacion.py
5	Theorem 1 in 16-dim	simple zero; $\tau = -12.84$; $a_3 = 5.09$ (det -11.67); real saddle-node	pieza1_teorema_4x4.py
6	Theorem 2 (cusp)	$a_2, a_3 \sim 10^{-11}$; versal = 1.30; $3/2$ law: $\propto (-a_2)^{1.503}$, $R^2 = 1.0$	pieza1_cuspide_codim2.py
7	invariants: $\text{tr } \Gamma^3$ outside the bilateral ring; Lemma 1	$\text{tr } \Gamma^3$ not G -invariant; real spectrum in metric-gradient	pieza1_no_hopf_invariantes.py
8	robustness / structural stability	(β, b_6, J) : 40/40 retain simple soft mode, fold, and det contribution	pieza1_robustez_teorema.py

The numerical study of the inertial and chaotic regime is in the companion document.

9. Discussion

When the nonlinearity of a matrix gradient flow with bilateral symmetry is fixed by that symmetry, the orientation-sensitive invariant, the determinant, determines the type of soft-mode bifurcation through its cofactor. That $\text{tr } \Gamma^3$ lies outside the bilateral ring is what distinguishes the determinant without recourse to a choice of potential: P uses the admitted generators of lowest degree. The determinant enters as the invariant that generates the orientation part of the cubic, not as an indicator of stability; that is fixed by the spectrum of the symmetric Hessian. Lemma 1 delimits the scope of the gradient sector.

On the $3/2$ exponent. The window of three equilibria of the cusp scales as $(-a_2)^{3/2}$. This is the universal law of the A_3 catastrophe, shared by any cusp, and by itself does not validate the determinant mechanism; what it validates is that the organizer is a genuine cusp. The measurable number is the signature of the class of bifurcation, not of the invariant that originates it.

Where these flows appear. Real matrix gradient flows arise in the Landau–de Gennes Q -tensor (§7), continuum strain tensors, and the loss landscapes of small neural networks, where an invariant nonlinearity governs the soft mode. In adversarial training (GANs) the gradient symmetry is broken natively by the min-max structure, exciting the oscillatory modes that Lemma 1 attributes to the non-gradient sector.

Scope note. We do not claim this Lagrangian is privileged: it is one of the class with bilateral symmetry, built from the admitted generators of lowest degree. The word “universal” is used only in the technical sense of universal unfolding of normal forms. The determinant generates the orientation part of the cubic within this class, not as a statement about arbitrary matrix flows.

Directions, treated in the companion document as exploration and not as theorems: the second-order inertial extension, where the Jordan block of Bogdanov–Takens is released (obstructed in the gradient limit by Lemma 1), with its coefficient certificate in $16 + 16$ dimensions open; the phenomenology of chaos (saddle-focus, Shilnikov) of the non-gradient sector; and the extension to $M_n(\mathbb{R})$, $n > 4$.

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